

Animal Cells

Cells, tissues, organs and systems

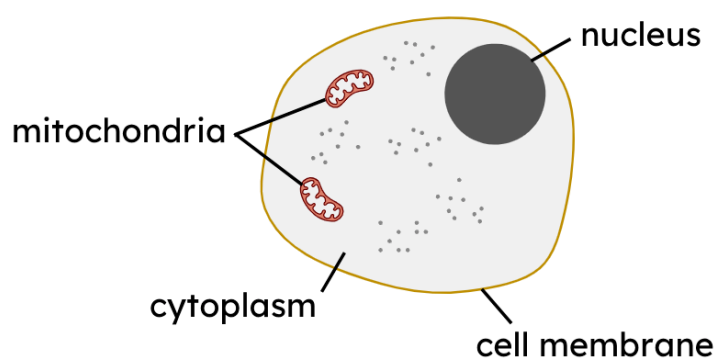
KEY INFORMATION

Cell membrane: The cell membrane surrounds the cell and controls what substances enter and leave the cell.

Cytoplasm: The cytoplasm is the jelly-like substance inside a cell where the chemical reactions of the cell take place.

Nucleus: The nucleus contains DNA, which stores information that controls the activities of the cell.

Mitochondria: Mitochondria use food and oxygen as fuel to provide energy for the cell in a process called cellular respiration.



A diagram of an animal cell

1. Match each term to its correct definition.

[4 marks]

Cell membrane

Cytoplasm

Nucleus

Mitochondria

Use food and oxygen to provide energy for the cell through cellular respiration

Surrounds the cell and controls what substances enter and leave

Jelly-like substance inside the cell where chemical reactions take place

Contains DNA, which stores information that controls the activities of the cell

2. Fill in the blanks to describe the **nucleus**

[2 marks]

The nucleus is a structure found in animal cells. It contains _____, which stores _____ that controls the activities of the cell.

3. Fill in the blanks to describe the **cell membrane** [2 marks]

The cell membrane surrounds the cell and controls what substances _____ and _____ the cell.

4. Fill in the blanks to describe **cytoplasm** [2 marks]

The cytoplasm is a _____-like substance found inside the cell. It is where the _____ of the cell take place.

5. Fill in the blanks to describe **mitochondria** [4 marks]

Mitochondria use _____ and _____ as fuel to provide _____ for the cell. This process is called cellular _____.

6. Name the part of an animal cell described in each of the following statements.

(a) This jelly-like substance fills the cell and is where the chemical reactions of the cell take place. [1 mark]

(b) This structure surrounds the cell and controls what substances enter and leave. [1 mark]

(c) This structure contains DNA, which stores information that controls the activities of the cell. [1 mark]

(d) These structures use food and oxygen as fuel to provide the cell with energy through cellular respiration. [1 mark]

7. State the function of each of the following animal cell structures.

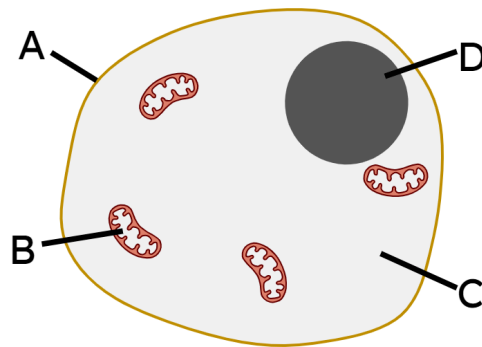
(a) Cell membrane [1 mark]

(b) Cytoplasm [1 mark]

(c) Nucleus [1 mark]

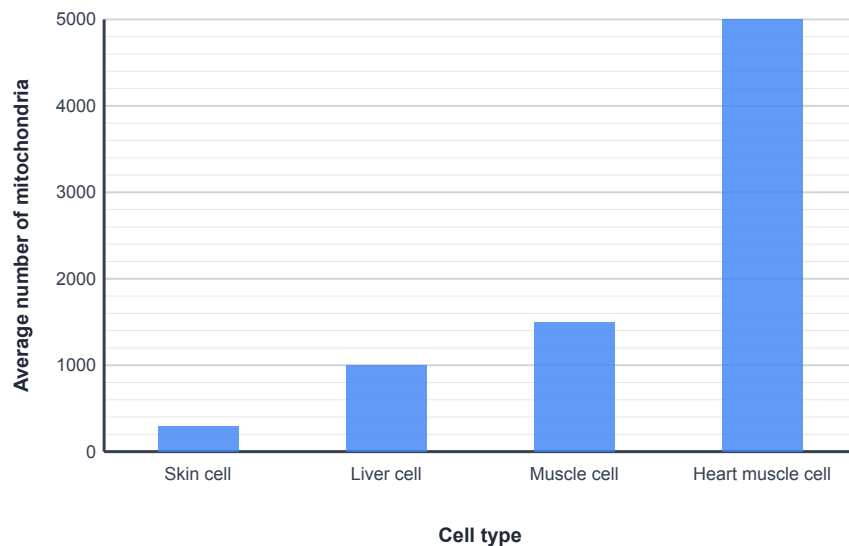
(d) Mitochondria [1 mark]

8. A scientist is studying muscle cells from a frog's leg. Muscle cells need a lot of energy to contract and move the frog's leg. The diagram below shows an animal cell.



- (a) Name the structures labelled A, B, C and D in the diagram. [4 marks]
- (b) Explain why a frog's muscle cells would be expected to contain a much greater number of mitochondria than the cells lining the inside of its mouth. [2 marks]
- (c) The nucleus controls the activities of the cell. Suggest what would happen to a frog's muscle cell if its nucleus was removed. Give a reason for your answer. [2 marks]
9. The bar chart shows the average number of mitochondria found in different types of animal cells. Use the bar chart to answer the questions below.

BAR CHART: AVERAGE NUMBER OF MITOCHONDRIA PER CELL TYPE



- (a) State which cell type contains the greatest number of mitochondria. [1 mark]
- (b) Calculate the difference in the average number of mitochondria between a liver cell and a skin cell. [1 mark]
- (c) Explain why heart muscle cells contain so many more mitochondria than skin cells. Use your knowledge of the function of mitochondria in your answer. [3 marks]

10. A muscle cell requires a constant supply of glucose and oxygen to function, and must also remove waste products such as carbon dioxide.

(a) State which part of the cell is responsible for allowing these substances to enter and leave? [1 mark]

(b) Which part of the cell does oxygen and glucose need to get to and for what process? [2 marks]

(c) Which part of the cell stores the information about which substances should be let in and out of the cell? [1 mark]

11. The table shows the typical sizes of five different animal cell types. Use the data in the table to answer the questions below.

GRAPH 1: SIZES OF DIFFERENT ANIMAL CELL TYPES

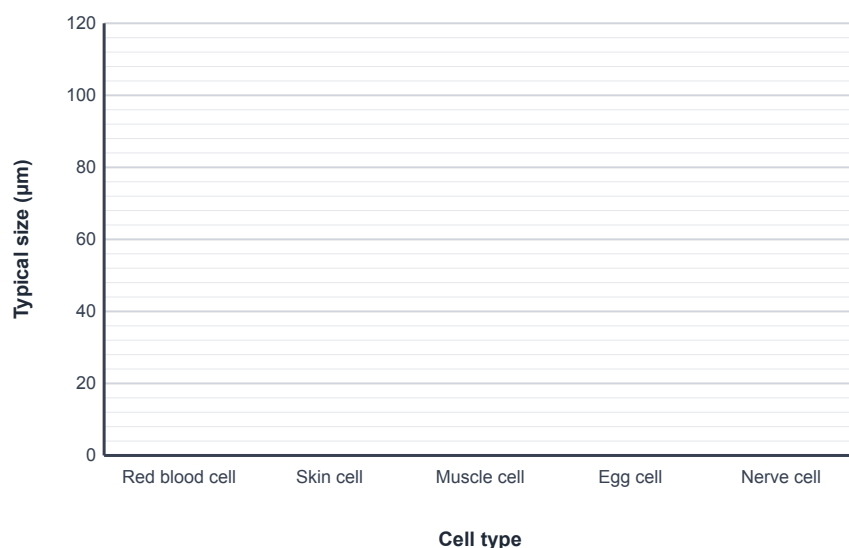


TABLE 1: SIZES OF DIFFERENT ANIMAL CELL TYPES

Cell type	Typical size (µm)
Red blood cell	8
Skin cell	30
Muscle cell	100
Egg cell	120
Nerve cell	20

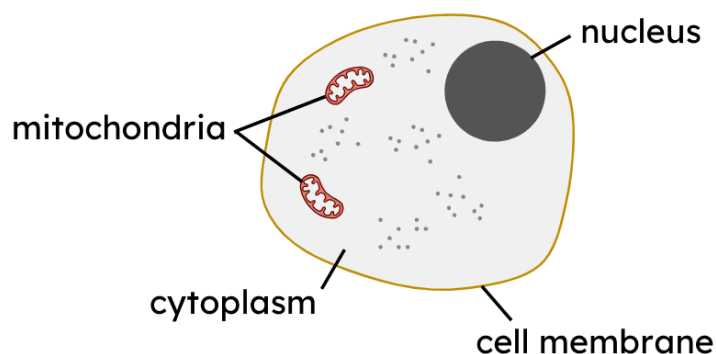
(a) Plot a bar chart of the data from the table onto the grid provided. Label each bar with the cell type and draw each bar to the correct height. [3 marks]

(b) State which cell type is the largest. [1 mark]

(c) Calculate the difference in size between the largest and smallest cell types shown in the table. [1 mark]

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A diagram of an animal cell

1. Match each term to its correct definition.

Cell membrane	Use food and oxygen to provide energy for the cell through cellular respiration
Cytoplasm	Surrounds the cell and controls what substances enter and leave
Nucleus	Jelly-like substance inside the cell where chemical reactions take place
Mitochondria	Contains DNA, which stores information that controls the activities of the cell

2. Fill in the blanks to describe the **nucleus**

The nucleus is a structure found in animal cells. It contains **DNA**, which stores **information** that controls the activities of the cell.

3. Fill in the blanks to describe the **cell membrane**

The cell membrane surrounds the cell and controls what substances **enter** and **leave** the cell.

4. Fill in the blanks to describe **cytoplasm**

The cytoplasm is a **jelly**-like substance found inside the cell. It is where the **chemical reactions** of the cell take place.

5. Fill in the blanks to describe mitochondria

Mitochondria use **food** and **oxygen** as fuel to provide **energy** for the cell. This process is called cellular **respiration**.

6. Name the part of an animal cell described in each of the following statements.

- (a)** This jelly-like substance fills the cell and is where the chemical reactions of the cell take place. [1 mark]

Cytoplasm [1]

- (b)** This structure surrounds the cell and controls what substances enter and leave. [1 mark]

Cell membrane [1]

- (c)** This structure contains DNA, which stores information that controls the activities of the cell. [1 mark]

Nucleus [1]

- (d)** These structures use food and oxygen as fuel to provide the cell with energy through cellular respiration. [1 mark]

Mitochondria [1]

7. State the function of each of the following animal cell structures.

- (a)** Cell membrane [1 mark]

The cell membrane surrounds the cell and controls what substances enter and leave the cell. [1]

- (b)** Cytoplasm [1 mark]

The cytoplasm is the jelly-like substance inside the cell where chemical reactions of the cell take place. [1]

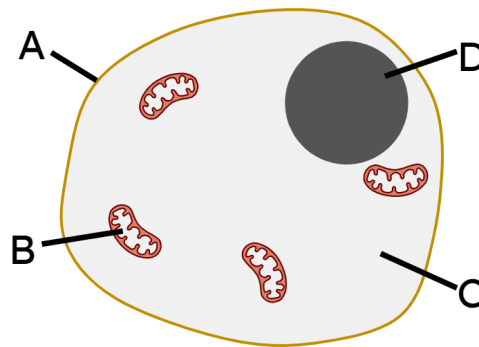
- (c)** Nucleus [1 mark]

The nucleus contains DNA, which stores information that controls the activities of the cell. [1]

- (d)** Mitochondria [1 mark]

Mitochondria use food and oxygen as fuel to provide energy for the cell through cellular respiration. [1]

8. A scientist is studying muscle cells from a frog's leg. Muscle cells need a lot of energy to contract and move the frog's leg. The diagram below shows an animal cell.



- (a) Name the structures labelled A, B, C and D in the diagram. [4 marks]

A = cell membrane [1]; B = mitochondria [1]; C = cytoplasm [1]; D = nucleus [1]

- (b) Explain why a frog's muscle cells would be expected to contain a much greater number of mitochondria than the cells lining the inside of its mouth. [2 marks]

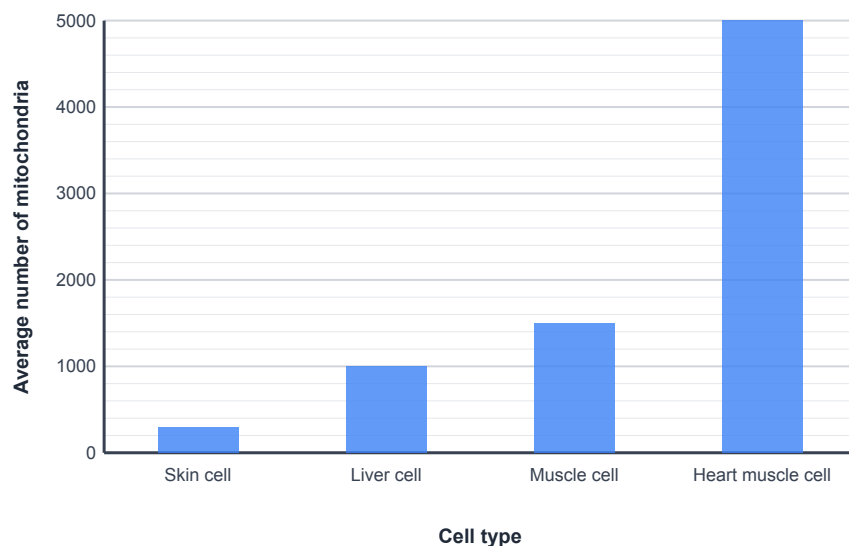
Mitochondria carry out cellular respiration, releasing energy from food and oxygen [1]; muscle cells need more energy for contraction / movement, so they require more mitochondria than less active cells [1]

- (c) The nucleus controls the activities of the cell. Suggest what would happen to a frog's muscle cell if its nucleus was removed. Give a reason for your answer. [2 marks]

The cell would not be able to function / carry out its normal activities correctly [1] because the nucleus contains DNA which stores the information needed to control the cell's activities [1] — accept any one valid point for 1 mark

9. The bar chart shows the average number of mitochondria found in different types of animal cells. Use the bar chart to answer the questions below.

BAR CHART: AVERAGE NUMBER OF MITOCHONDRIA PER CELL TYPE



- (a) State which cell type contains the greatest number of mitochondria. [1 mark]

Heart muscle cell [1]

Question 9 (continued)

- (b)** Calculate the difference in the average number of mitochondria between a liver cell and a skin cell. [1 mark]

1000 – 300 = 700 [1]

- (c)** Explain why heart muscle cells contain so many more mitochondria than skin cells. Use your knowledge of the function of mitochondria in your answer. [3 marks]

Mitochondria carry out cellular respiration / provide energy for the cell [1]; heart muscle cells need to contract continuously / are constantly active [1]; therefore they need much more energy than skin cells, which are less active [1].

10. A muscle cell requires a constant supply of glucose and oxygen to function, and must also remove waste products such as carbon dioxide.

- (a)** State which part of the cell is responsible for allowing these substances to enter and leave? [1 mark]

Cell membrane

- (b)** Which part of the cell does oxygen and glucose need to get to and for what process? [2 marks]

Mitochondria[1] for respiration [1]

- (c)** Which part of the cell stores the information about which substances should be let in and out of the cell? [1 mark]

Nucleus [1]

11. The table shows the typical sizes of five different animal cell types. Use the data in the table to answer the questions below.

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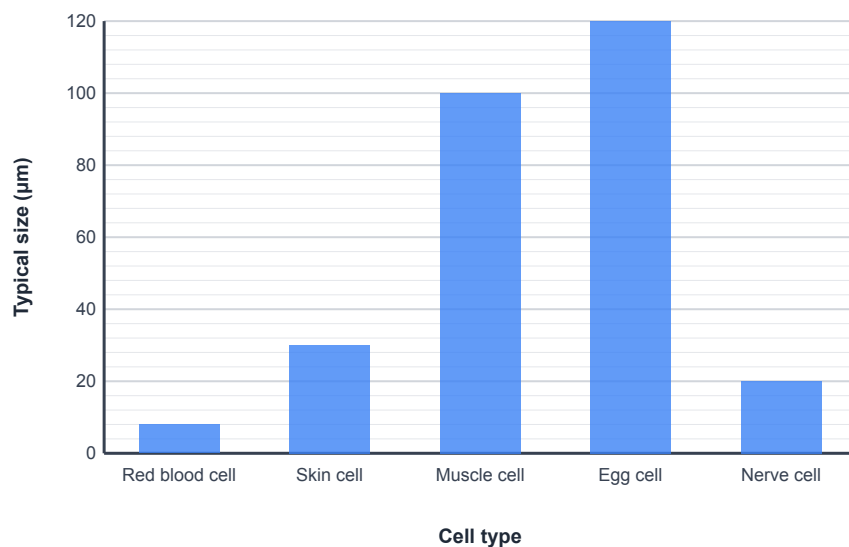


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Cell type	Typical size (µm)
Red blood cell	8
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Nerve cell	20

- (a) Plot a bar chart of the data from the table onto the grid provided. Label each bar with the cell type and draw each bar to the correct height. [3 marks]

Award 1 mark for all five bars drawn at correct heights (8, 30, 100, 120, 20 µm); 1 mark for all bars correctly labelled with cell type; 1 mark for y-axis labelled with units (µm) and a sensible scale used. [3]

- (b) State which cell type is the largest. [1 mark]

Egg cell [1]

- (c) Calculate the difference in size between the largest and smallest cell types shown in the table. [1 mark]

$120 - 8 = 112 \text{ µm}$ [1]